

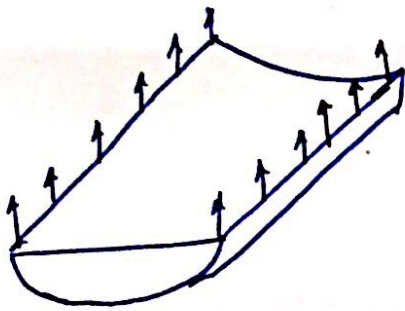
Thin Cylindrical shell of Circular Cross Section:-

In this topic analysis of thin cylindrical shells are taken up. When the thickness of the cylindrical is less than $1/10^{\text{th}}$ of its radius. Then we consider the structural component to be a shell and analyse the same as a shell.

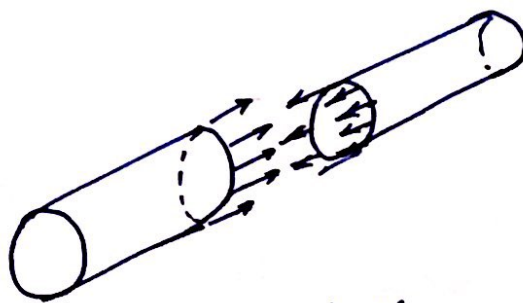
Stresses in thin cylinders.

In general three types of stresses are developed,

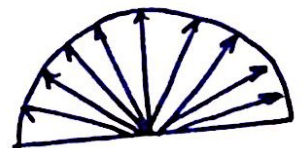
- (a) Circumferential stresses or otherwise called hoop stresses
- (b) longitudinal stresses and
- (c) Radial stresses.



Hoop stress.

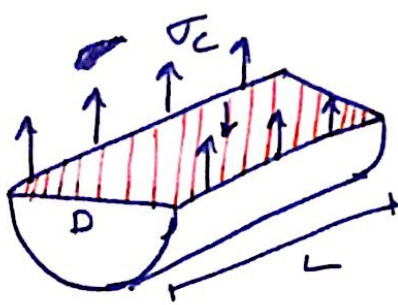


Longitudinal stress.



Radial stresses.

In a thin walled cylinder the circumferential stresses may be constant over the wall thickness. Stresses in the radial directions may be neglected as it is very small relatively.



(2)

Bursting force = $P \times d \times L$ - (1)
acting over the area in red.

Resisting force = $\sigma_c \times t \times L \times 2$ - (2)

where t is the thickness of the cylinder.

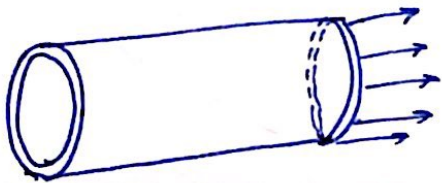
Under equilibrium conditions. Equation (1) should be equal to (2).

$$P d L = 2 \sigma_c t \times L$$

$$\sigma_c = \frac{P d}{2 t} \text{ after rearranging.}$$

The above relation is for hoop stress.

Longitudinal stress.



$$\text{The bursting force} = P \times \frac{d^2 \pi}{4} \quad - (1)$$

and the resisting force is given by $\sigma_L \times t \times \pi \times d$ - (2)

Under equilibrium conditions, (1) = (2).

$$\sigma_L \times t \times \pi d = P \times \pi d^2 / 4$$

Rearranging the terms,

$$\sigma_L = \frac{P d}{4 t} \quad // \quad \text{This is the relation for longitudinal stress.}$$

Hence, we can say $\sigma_c = 2 \sigma_L$

Changes in dimensions of the Cylinder:-

(3)

$$\left. \begin{aligned} \sigma_c &= Pd/2t \\ \sigma_L &= Pd/4t \end{aligned} \right\} \text{ Assuming poisson's ratio as } \mu \text{ and } \text{youngs modulus as } E, \text{ we get,}$$

$$E_c = \frac{\sigma_c}{E} - \mu \frac{\sigma_L}{E}$$

$$E_c = \frac{1}{E} [Pd/2t - \mu Pd/4t] = Pd/4tE [2 - \mu]$$

If δd be the change in diameter then,

$$E_c = \text{Circumferential strain} = \delta d/d$$

$E_c = \text{Change in diameter / Original diameter.}$

$$E_c = \delta d/d = Pd/4tE [2 - \mu]$$

Longitudinal strain:- $\delta L/L = E_L = \frac{\sigma_L}{E} - \mu \frac{\sigma_c}{E}$

$$= \frac{1}{E} [Pd/4t - \mu Pd/2t]$$
$$E_L = \frac{Pd}{4tE} [1 - 2\mu].$$

Change in Volume:- $V = \pi d^2/4 \times L$ which is the original volume.

As V is both a function of L and d , after ~~small~~ change in volume,

$$dV = \frac{\pi}{4} 2d \delta d L + \frac{\pi}{4} d^2 \delta L, \quad dV = \frac{\pi d^2}{4} L \left[\frac{2\delta d}{d} + \frac{\delta L}{L} \right]$$

$$\frac{dV}{V} = 2 \frac{\delta d}{d} + \frac{\delta L}{L} \quad \text{which can be re-written as,}$$

$$\frac{dV}{V} = 2E_c + E_L //$$

Thin Spherical Shells:-

(4)



Consider the hemi-sphere as shown,
the bursting force within the sphere would be
the internal pressure acting over the area $\frac{\pi d^2}{4}$

Hence the bursting force would be
$$= P \times \frac{\pi d^2}{4} \quad \text{--- (1)}$$

and the resisting force would be,

$$\pi d \times t \times \sigma \quad \text{--- (2)}$$

Equating (1) and (2) Under equilibrium conditions,

$$P \times \frac{\pi d^2}{4} = \pi d \times t \times \sigma$$

$$\sigma = \frac{Pd}{4t}$$

The longitudinal and Circumferential are the same.

$$\sigma_c = \sigma_L = Pd/4t.$$

Change in dimensions, Circumferential strain will be equal
to longitudinal strain

$$\epsilon = \frac{\sigma_c}{E} - \mu \frac{\sigma_L}{E}$$

$$\epsilon = \frac{Pd}{4tE} [1 - \mu] \quad //$$

$$\text{Volume of the spherical shell} = \frac{4}{3} \pi r^3$$

$$\text{or } \frac{\pi d^3}{6}$$

$$\delta V = \frac{\pi}{6} 3d^2 \delta d$$

$$\frac{dV}{V} = \frac{\frac{\pi}{6} 3d^2 \delta d}{\frac{\pi d^3}{6}} = \frac{3\delta d}{d} = 3\epsilon$$

$$\frac{dV}{V} = \frac{3Pd}{4tE} [1-\mu]. \quad \text{, This is the relationship for the Volumetric strain of a spherical shell.}$$

Cylindrical shells with Hemi-spherical Ends:-

$$\text{Hoop stress for the cylindrical shells} = \frac{Pd}{2t}$$

$$\text{Longitudinal stress for the cylindrical shells} = \frac{Pd}{4t}$$

Stresses in the spherical / hemi-spherical ends,

$$\text{Hoop stress} = \text{longitudinal stress,} = Pd/4t$$

$$\text{Hoop strain} = \frac{Pd}{4tE} [1-\mu] \quad \text{--- (1)}$$

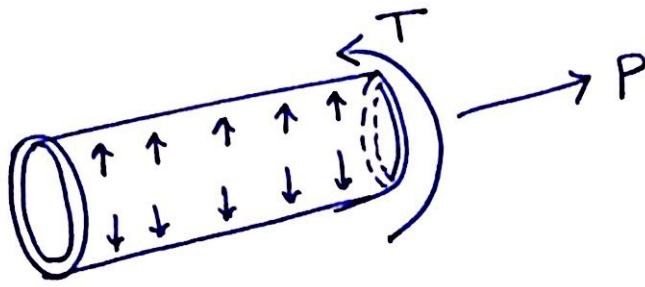
$$\text{And hoop strain for cylinder} = \frac{Pd}{4tE} [2-\mu] \quad \text{--- (2)}$$

Equating (1) and (2). if there is a change in thickness then the ratio will be

$$t_c/t_s = \frac{0.7}{1.7} \quad //$$

Shells Under Combined loading:-

6



Consider a cylindrical shell under internal pressure, and an external axial pull and Torsion.

Now, due to complex system of loading we can calculate principle stresses.

Due to internal pressure, we will have hoop stress and longitudinal stress.

Hence, stresses in the x-direction will be a addition of longitudinal stress and stress due to axial pull.

$$\sigma_x = \sigma_L + \sigma_P = \frac{Pd}{4t} + P_A/A.$$

$$\text{Similarly, } \sigma_y = \text{hoop stress} = \frac{Pd}{2t}$$

τ_{xy} is the shear stress developed in the shell due to Torsion T.

Hence, the principle strains developed in is given by.

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} //$$